BUSINESS REQUIREMENTS DOCUMENT

E-Commerce Checkout Flow — Cart to Order Tracking Handoff

Field	Detail
Document Owner	Harshal Sali (Business Analyst — Portfolio Project)
Platform Context	AEM-integrated commerce platform (Magento backend), B2B + B2C — modeled on real-world implementation patterns
Version	1.0
Status	Draft for Review
Date	25 July 2026

1. Purpose & Overview

This document defines the business and functional requirements for the end-to-end checkout experience — from cart review through payment, order confirmation, and handoff to order tracking — for a B2B/B2C e-commerce platform. It is intended to align business stakeholders, UX, and engineering on scope, rules, and acceptance criteria prior to development.

2. Business Objectives

- Reduce cart abandonment by minimizing checkout friction (guest checkout, clear error handling, transparent pricing)
- Support both B2C card-based payments and B2B account/PO-based payments within a single unified flow
- Ensure checkout is accurate and compliant (tax handling, PCI compliance) before scaling promotional campaigns
- Provide customers visibility into order status immediately after purchase to reduce post-purchase support inquiries

3. Scope

In Scope

- Cart review, shipping address entry & validation, shipping method selection
- Promo code / discount application
- Payment: credit/debit card, B2B PO number, B2B approved credit account
- Order review, confirmation, and initial handoff to order tracking status

Out of Scope

- Post-purchase returns/refunds processing (covered in a separate BRD)
- Loyalty/rewards points redemption
- Multi-currency checkout (single-currency, US market only, for this release)

4. Stakeholders

Role	Interest
Business Sponsor / E-commerce Manager	Conversion rate, revenue, cart abandonment reduction
B2B Sales/Accounts Team	PO and credit account payment accuracy for enterprise clients
Finance / Tax Team	Correct tax calculation and exemption handling
Customer Support	Reduced post-purchase inquiries via clear confirmation and tracking
Engineering / Platform Team	Feasibility and technical constraints of AEM/Magento

	integration
QA Team	Testability of business rules and edge cases

5. Assumptions & Constraints

- Payment gateway integration (tokenized card processing) is handled by an existing certified third-party provider; this BRD does not define gateway internals.
- B2B account approval and credit line status are managed upstream in the CRM/ERP system and passed to the commerce platform via existing integration.
- Single-market (US) release; international tax/shipping rules are out of scope.

6. Checkout Process Flow

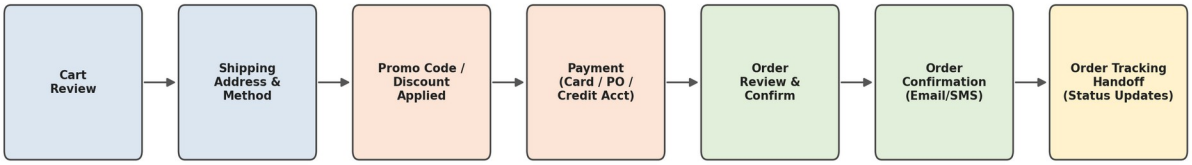


Figure 1: End-to-end checkout flow from cart review to order tracking handoff.

7. User Stories & Priority

ID	Title	User Story	Priority
US-01	Guest checkout	As a guest shopper, I want to complete checkout without creating an account, so that I can purchase quickly without friction.	High
US-02	B2B PO / credit account payment	As a registered B2B customer, I want to pay via company PO number or approved credit account, so that I can complete bulk orders per my organization's procurement policy.	High
US-03	Apply promo code	As a shopper, I want to apply a promo code at checkout, so that I receive an eligible discount before payment.	Medium
US-04	Shipping method selection	As a shopper, I want to choose from available shipping methods with visible cost and delivery estimate, so that I can select based on urgency and budget.	High
US-05	Address validation	As a shopper, I want my shipping address validated in real time, so that my order isn't delayed or misdelivered due to an	High

		invalid address.	
US-06	Order review before payment	As a shopper, I want to review a full order summary (items, tax, shipping, discounts) before submitting payment, so that I can verify accuracy.	High
US-07	Failed payment handling	As a shopper, I want a clear, specific error message if my payment fails, so that I can correct the issue and retry without abandoning my cart.	High
US-08	Order confirmation	As a shopper, I want to receive an order confirmation via email/SMS immediately after purchase, so that I have proof of order and a reference number.	High
US-09	Tax exemption (B2B)	As a tax-exempt B2B customer, I want applicable taxes automatically excluded at checkout based on my account status, so that I'm not incorrectly charged.	Medium
US-10	Order tracking handoff	As a shopper, I want to view real-time order status after checkout, so that I know when to expect delivery.	Medium

8. Acceptance Criteria (Sample — Given/When/Then)

US-01: Guest Checkout

- Given a shopper has items in cart and is not logged in, When they proceed to checkout, Then they are offered a "Checkout as Guest" option without forced registration
- Given a guest completes checkout, When the order is placed, Then an order confirmation is sent to the email provided, with an option to create an account post-purchase

US-02: B2B PO / Credit Account Payment

- Given a logged-in B2B account with an approved credit line, When they reach the payment step, Then "Pay via PO Number" and "Pay via Credit Account" are available alongside card payment
- Given a B2B account without an approved credit line, When they reach the payment step, Then only card payment is displayed

US-07: Failed Payment Handling

- Given a shopper submits payment, When the payment gateway declines the transaction, Then a specific, actionable error message is displayed (e.g., "card declined — please verify details or use another payment method"), and cart contents are preserved

9. Business Rules

- Promo codes cannot be combined unless explicitly flagged as "stackable" in the promotion configuration.

- B2B credit account / PO payment is available only to accounts with an approved credit line status; all other accounts default to card payment only.
- Tax exemption is applied automatically only for accounts with a verified tax-exempt certificate on file; manual exemption requests are out of scope for this release.
- Free shipping threshold applies to merchandise subtotal only, excluding tax and existing discounts.
- Orders remain editable (address/shipping method) until payment authorization is confirmed; no edits permitted after authorization.
- Guest checkout orders can be converted to a registered account post-purchase but cannot access account-based pricing retroactively.

10. Non-Functional Requirements

Category	Requirement
Performance	Checkout page load ≤ 2.5s on 4G mobile; payment authorization response ≤ 5s under normal load.
Security	PCI-DSS compliant payment handling; no raw card data stored on application servers.
Availability	99.9% uptime target for checkout flow, including during peak promotional periods.
Compatibility	Fully functional across major browsers and on iOS/Android mobile and tablet, per BrowserStack test matrix.
Accessibility	WCAG 2.1 AA compliance for all checkout form fields and error messaging.

11. Risks & Dependencies

Risk / Dependency	Impact	Mitigation
Payment gateway downtime during peak sales	High — direct revenue loss	Fallback payment method + real-time monitoring alerts
CRM/ERP credit-status sync delay	Medium — B2B customers see incorrect payment options	Cache last-known status with defined refresh SLA; manual override for support team
Tax calculation service dependency	Medium — incorrect order totals	Contract-level SLA with tax service provider; fallback manual review queue for failures

12. Open Questions

- Should guest checkout orders above a certain value threshold require additional verification (fraud prevention)?
- What is the retry limit before a failed payment locks the checkout session, and what happens to reserved inventory during retries?
- Does the B2B PO payment flow require manager approval workflow within this release, or is it deferred to a future phase?

13. Glossary

- **PO:** Purchase Order — a B2B payment method referencing an internal procurement document.
- **UAT:** User Acceptance Testing — business-side validation prior to production release.
- **SLA:** Service Level Agreement — defined performance/response commitment.